

The Annotation Ceiling in Forensic Bruise Segmentation: When a Model Agrees with Expert Consensus Better Than the Expert It Learned From

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Abstract

White-light photography of bruises is central to forensic injury documentation, yet automatic bruise segmentation is trained and evaluated against annotations that expert clinicians themselves disagree about. We study this problem on the NIJ white-light bruise corpus under a deliberately strict, application-faithful protocol: models are trained on a single expert’s masks and evaluated on the *2-of-3 majority consensus* of the three highest-volume annotators, over the 185-image / 28-subject strict intersection that all three labeled. Under this cross-annotator generalization test we make three contributions. First, we measure the *annotation ceiling*: the three experts agree with one another at only 0.639 mean Dice, well below the agreement between our models and the consensus target. Second, and counterintuitively, a compact SegFormer-B0 trained *only* on one annotator (Paul) agrees with the three-expert consensus significantly better than that annotator does himself (mean Dice +0.067, subject-level cluster-bootstrap 95% CI [+0.003, +0.133]): training averages away an annotator’s idiosyncrasies and keeps what generalizes. Third, in a controlled comparison of seven architectures (SegFormer-B2/B0, YOLO26n-sem, U-Net, DeepLabV3+) trained under one recipe, every capable model clusters in a narrow 0.74–0.77 mean-Dice band; the differences that *are* resolvable at this consensus-limited sample size live in failure rates and cross-seed stability, not in headline Dice, and distillation is statistically indistinguishable from its baseline. Our results reframe “underpowered” from an apology into a mechanism: forensic bruise segmentation is *label-limited*, *not capacity-limited*, and the field’s binding constraint is annotation consensus, not model architecture.

1 Introduction

Bruises are among the most common findings in cases of interpersonal violence, elder abuse, and child maltreatment, and their photographic documentation carries evidentiary weight. Automatic segmentation of the bruised region from a clinical photograph promises consistent, reproducible documentation. In practice, however, the “ground truth” used to train and score such systems is produced by human annotators whose

delineations disagree substantially, because the boundary of a bruise is genuinely ambiguous: it fades gradually into normal skin and its apparent extent depends on lighting, body site, and the rater. This makes bruise segmentation an unusually honest testbed for a question that pervades applied medical imaging but is rarely confronted directly: *what can a segmentation benchmark actually tell us when the labels themselves are noisy and the consensus-labeled test set is necessarily small?*

We approach this as an application study on the NIJ white-light bruise corpus. Our design is driven by two facts about the data. (i) The corpus was labeled by many annotators; only the three highest-volume annotators (denoted Paul, Gbarimah, Erik) share enough images to define a common evaluation set. Their strict intersection is **185 images from 28 subjects**. (ii) There is no way to enlarge this consensus set without commissioning new expert annotation: $n = 28$ subjects is a *ceiling imposed by the availability of three-annotator consensus labels*, not a design choice. We therefore treat the majority vote of the three experts as the test target, train on the non-intersection images of a single, previously-selected best annotator (Paul), and evaluate every model as a *cross-annotator generalization* problem: how well does a model trained on one expert reproduce the consensus of three?

Within this protocol we compare seven segmentation models spanning three families—a semantic transformer (SegFormer-B2/B0), a detector-derived semantic head (YOLO26n-sem), and two classic encoder–decoder CNNs (U-Net, DeepLabV3+)—all trained under one shared recipe so that architecture is the isolated variable. Rather than crown a winner, we ask which conclusions survive the statistical reality of a 28-subject consensus test set, using subject-level cluster bootstrapping throughout.

Contributions.

- An application-faithful, multi-annotator evaluation protocol for white-light bruise segmentation: train on one expert, test on three-expert majority consensus over the strict labeled intersection, with subject-disjoint splits.
- A quantified *annotation ceiling*: expert–expert agreement (0.639 mean Dice) is lower than model–consensus

agreement, establishing that the models are label-limited.

- The main empirical finding: a model trained on a single annotator agrees with the three-expert consensus *significantly better than that annotator himself* (mean Dice +0.067, 95% CI [+0.003, +0.133]). To our knowledge this “model-beats-its-annotator” effect has not been reported for forensic imaging.
- A controlled seven-architecture comparison showing convergence at the ceiling: capable models differ in failure rate and stability, not headline Dice, and knowledge distillation is not resolvable from zero at this sample size.

2 Related Work

Medical and forensic image segmentation. Encoder-decoder CNNs (U-Net [2], DeepLabV3+ [3]) and their self-configuring descendants (nnU-Net [4]) remain the default for biomedical segmentation, while transformer segmenters such as SegFormer [1] have become strong, efficient alternatives. Recent one-stage detectors expose semantic-segmentation heads (e.g., YOLO26 [5]), enabling real-time masks. Forensic bruise imaging has historically focused on *visibility* rather than automatic delineation, with alternate-light and cross-polarized techniques studied for enhancing bruise contrast [9]. We contribute an application study that segments bruises on standard white-light photographs and, critically, evaluates against a multi-expert consensus rather than a single rater.

Label noise and inter-observer variability. That expert annotations disagree is well documented in medical imaging [8, 7], and consensus-fusion methods such as STAPLE [7] estimate a hidden “true” segmentation from multiple raters. Our aim is different: we use the observed multi-rater disagreement as a *measuring stick*—an annotation ceiling—and quantify how model-consensus agreement compares to expert-consensus agreement. The finding that a model can match consensus better than an individual rater is related to the “wisdom of crowds” rationale behind label fusion, but we show it emerges from *single-annotator* training via generalization, not from fusing raters.

Knowledge distillation. Knowledge distillation [6] transfers a large teacher’s soft predictions to a compact student and is widely used to shrink segmentation models. We evaluate it here as one of the interventions whose effect we test for statistical resolvability at forensic sample sizes.

Uncertainty at small sample sizes. When test images are clustered within few subjects, the effective sample size is the number of subjects, and subject-level (cluster) resampling is

required for valid inference [10]. We adopt paired subject-level cluster bootstrapping for all model comparisons, which is both correct for a shared test set and tighter than independent resampling.

3 Dataset and Multi-Annotator Design

Corpus and consensus set. We use the NIJ white-light bruise corpus. Among its annotators, the three with the largest labeled volume (Paul, Gbarimah, Erik) share a common set of images; their **strict intersection is 185 white-light images from 28 subjects**, which we fix as the test set. The test target is the **2-of-3 pixel majority vote** of the three experts—a pixel is positive iff at least two of the three annotators marked it positive.

Training data and disjointness. A prior five-fold cross-annotator study selected Paul as the most stable annotator for training. All models are therefore trained on Paul’s *non-intersection* images (697 train / 134 validation), which are *subject-disjoint* from the 28 test subjects by construction. Evaluating a Paul-trained model against the three-expert majority is thus a genuine cross-annotator generalization test, not a within-distribution split.

The $n = 28$ ceiling. There are 28 test subjects because that is how many subjects all three experts annotated; a 29th requires new expert annotation. This reframes every “underpowered” result: it is not that our study is too small, but that *at the maximum size for which three-annotator consensus exists, model-level differences are unresolvable*—a statement about the field, not the experiment.

4 Methods

Preprocessing. Every image and mask is resized once to a common 640×640 grid (bilinear for images, nearest for masks) so that prediction and ground truth are compared on one grid and no mask is ever resized to match another. Pixels are scaled to $[0, 1]$; the ImageNet normalization each backbone expects is applied inside the model, so a single neutral data pipeline feeds all architectures.

Architectures. We evaluate: SegFormer-B2 (teacher) and SegFormer-B0 (student) transformer segmenters; the YOLO26n semantic-segmentation head (YOLO26n-sem); and U-Net and DeepLabV3+ with ResNet-50 encoders as classic baselines. Parameter counts span 1.6M–32.5M (Table 1).

Training recipe. All non-detector models share one recipe: Dice+BCE loss, AdamW with a separate (lower) learning

rate for the pretrained encoder and a $10\times$ rate for the randomly initialized decoder, no weight decay on norm/bias parameters, linear warmup then polynomial decay, and early stopping on validation. YOLO26n-sem is trained with its native detector pipeline (its own letterbox preprocessing, augmentation, and schedule), and evaluated with its native argmax decision, which is how it was designed to run. SegFormer models are trained with three seeds; U-Net/DeepLabV3+ baselines with a single seed (a stated limitation).

Knowledge distillation. For the distilled variants, the SegFormer-B0 student is trained with an additional term matching the SegFormer-B2 teacher’s soft outputs; the YOLO distilled variant uses offline teacher pseudo-masks. Distillation weights follow the project’s tuned values.

Operating point. A model emits a per-pixel score; a binary mask requires a threshold. For each model we sweep the decision threshold *on the validation set only* and apply the chosen value once to the test set. Thresholding a monotone score is equivalent to thresholding the raw logit, so the choice is a single one-dimensional cut and never touches the test data.

Metrics and inference. We report per-image Dice and IoU, and—separately—the *complete-miss rate*: the fraction of images that contain a bruise but for which the model predicts no positive pixels. A blank mask is a missed injury and is far worse than a loose outline, so it is reported on its own. Because per-image Dice is strongly bimodal (a model either localizes the bruise or misses it), we emphasize the median and use percentile bootstraps rather than normal-theory intervals. All comparisons use paired, subject-level cluster bootstrapping ($B=4000$ over the 28 subjects).

5 Results

The models cluster; the winner depends on the metric.

Table 1 summarizes test performance. The three SegFormer variants and the two classic CNN baselines all sit within a 0.74–0.77 mean-Dice band and a 0.80–0.83 median-Dice band. Strikingly, the *highest median Dice* (0.828) belongs to U-Net, whose complete-miss rate (5.4%) is an order of magnitude worse than SegFormer’s ($\leq 0.5\%$): the best-Dice model is not the safest model. The compact SegFormer-B0 (3.7M parameters) matches or beats both classic baselines (32.5M, 26.7M) at roughly one-eighth the parameters and near-zero miss rate—a clean accuracy/efficiency result. YOLO26n-sem is the fastest by far (122 FPS) but the least accurate and, importantly, the least stable (cross-seed std up to 0.065 vs. ≤ 0.006 for SegFormer), with one distilled seed collapsing entirely.

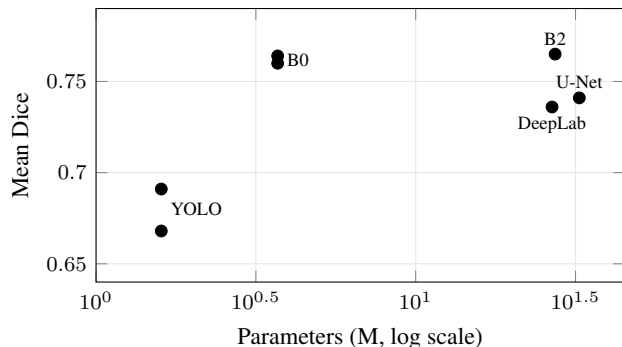


Figure 1: Accuracy vs. model size. SegFormer-B0 (3.7M) sits on the accuracy frontier, matching the $\sim 9\times$ larger classic baselines.

Distillation is not resolvable from zero. The distilled SegFormer-B0 improves mean Dice by only $+0.004$ over its direct counterpart—smaller than the ± 0.005 cross-seed standard deviation—and a paired subject-level cluster bootstrap does not separate the effect from zero. For YOLO, distillation *hurts* (mean 0.668 vs. 0.691) and inflates variance. At this consensus-limited sample size, distillation is a wash.

Efficiency–accuracy trade-off. Figure 1 plots mean Dice against parameter count. SegFormer-B0 dominates the classic baselines (higher Dice, $\sim 9\times$ fewer parameters); the teacher buys no measurable accuracy over the distilled student; YOLO trades a large accuracy and safety penalty for throughput.

6 The Annotation Ceiling

The experts barely agree with each other. Table 2 reports pairwise inter-annotator Dice on the 185 consensus images, computed by pure mask arithmetic (no model). The three experts agree with one another at only 0.639 mean Dice on average. Every capable model in Table 1 agrees with the majority target *better* than the experts agree with each other. The labels are noisier than the models.

Paul is the outlier we train on. Gbarimah and Erik agree with each other most (0.755), so the majority vote is effectively “what Gbarimah and Erik agreed on.” Paul—whose masks we train on—matches that target *least* of the three (0.700 vs. 0.873, 0.866). This does not contradict Paul’s selection for training, which was based on the downstream generalization of a Paul-trained model, a different criterion from agreeing with peers.

Main result: the model beats the annotator it learned from. We compare the val-selected SegFormer-B0 (distilled) against Paul, both scored against the same three-expert majority, with a paired subject-level cluster bootstrap ($B=4000$, 28 subjects). The model agrees with the consensus

Table 1: Test performance on the 185-image / 28-subject consensus set (majority target, scored at 640×640). SegFormer/YOLO are mean±std over three seeds (YOLO uses native argmax); U-Net/DeepLabV3+ are single-seed (†). Speed is single-frame latency on an A100. Every capable model falls in a narrow 0.74–0.77 mean-Dice band; the spread is in complete-miss rate and stability.

Model	Params (M)	Mean Dice	Median Dice	Complete-miss (%)	FPS (A100)
SegFormer-B2 (teacher)	27.3	0.765 ± 0.004	0.811	0.00	29.7
SegFormer-B0 (direct)	3.7	0.760 ± 0.006	0.811	0.54	60.0
SegFormer-B0 (distilled)	3.7	0.764 ± 0.005	0.811	0.18	60.1
U-Net (ResNet-50) [†]	32.5	0.741	0.828	5.41	—
DeepLabV3+ (ResNet-50) [†]	26.7	0.736	0.800	2.16	—
YOLO26n-sem (direct)	1.6	0.691 ± 0.020	0.802	7.57	122.2
YOLO26n-sem (distilled)	1.6	0.668 ± 0.065	0.766	7.57	121.7

Table 2: Inter-annotator agreement on the 185 consensus images. The three experts agree with each other far less (0.639 mean) than models agree with the majority target. Paul—the annotator we train on—agrees with the majority *least* of the three.

Comparison	Mean Dice	Median Dice
Gbarimah ↔ Erik	0.755	0.809
Paul ↔ Gbarimah	0.581	0.633
Paul ↔ Erik	0.581	0.616
Avg. human–human	0.639	—
Gbarimah vs. majority	0.873	0.926
Erik vs. majority	0.866	0.892
Paul vs. majority	0.700	0.750

significantly better than Paul does:

$$\Delta_{\text{mean}} = +0.067, \text{ CI } [+0.003, +0.133], P(\Delta > 0) = 98.0\%,$$

$$\Delta_{\text{median}} = +0.063, \text{ CI } [-0.002, +0.129], P(\Delta > 0) = 96.9\%.$$

(all Dice; 95% subject-level cluster-bootstrap CIs.) A model trained *only* on Paul’s masks reproduces the three-expert consensus better than Paul himself: training averages away one annotator’s idiosyncrasies and retains what generalizes. For scale, this effect is an order of magnitude larger than—and unlike—the distillation effect (+0.004, n.s.) that a conventional benchmark would have foregrounded.

Why this explains the null model comparisons. We were attempting to resolve ~ 0.02 -Dice architecture differences against a target that carries roughly 0.36 of annotation noise ($1 - 0.639$). That was never going to resolve. The models are label-limited, not capacity-limited; the architecture ranking is inside the noise floor, while the label-versus-model gap is not.

7 Discussion

Our results invert the usual reading of a flat leaderboard. The convergence of five capable architectures into a narrow Dice band is not a failure to find a better model; it is evidence that model capacity is no longer the binding constraint for this task at its current label quality. Two consequences follow for practitioners. First, *report failure rates*

and cluster-level intervals, not just mean Dice: complete-miss rate cleanly separates the models (0–0.5% for SegFormer, 2.2% for DeepLabV3+, 5.4% for U-Net, 7.6% for YOLO) where Dice does not, and it is the clinically decisive quantity. Second, *invest in annotation, not architecture:* at 28 subjects the minimum resolvable median-Dice effect is ~ 0.04 , larger than the gaps between transformer, CNN, and detector families, so further architecture search on this target buys unmeasurable gains, whereas additional consensus annotation directly raises the ceiling. The “model beats its annotator” effect suggests a concrete path: models trained on individual raters can serve as consensus-approximating pre-labelers to accelerate the very annotation the field is short of.

8 Limitations

The consensus test set is fixed at 28 subjects; this is a ceiling, but it bounds statistical power and we do not claim architecture rankings beyond it. The U-Net/DeepLabV3+ baselines are single-seed and used a marginally different train/validation partition (identical test set), so their point estimates lack the seed error bars of the transformer models; a matched multi-seed re-run is planned. YOLO is trained and scored in its native pipeline rather than the shared recipe, which is the fair choice for a detector but means its numbers reflect a different preprocessing chain. Intervals capture test-set sampling only, not training stochasticity, so they are, if anything, optimistic. Finally, we deliberately omit skin-tone subgroup (fairness) analysis: the worst-group Dice gap is an extremum-of-extrema statistic whose subject-level cluster-bootstrap interval at this sample size is too wide to support a claim, and reporting it as a finding would repeat exactly the over-interpretation this paper cautions against.

9 Conclusion

On a strict, multi-annotator forensic bruise benchmark we show that expert annotators agree with one another less than our models agree with their consensus, and that a compact model trained on a single annotator reproduces three-expert consensus significantly better than that annotator himself.

Across seven architectures, headline Dice is unresolvable at the consensus-limited sample size, while failure rate and stability are not. Forensic bruise segmentation is label-limited: the path forward is more consensus annotation and honest, cluster-level reporting—not a larger model.

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